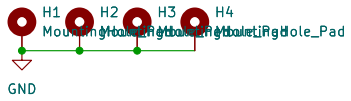
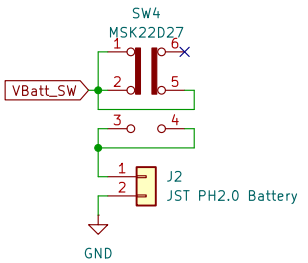
Note about 5V Power:
– When USB is connected this will be 5V from VBUS
– When running from a battery this will be Vbatt–V_protection_diode
– Battery can only charge when USB is connected and power switch is 'ON'
Notes about GPIOs:
– ADC: GPIO1 to 10 are available as ADC0 to 9
– GPIO3 is overloaded with JTAG src
– GPIO45, 46 are bootstrapping pins
Rev 1
– OSHpark test release 10/27/2026
Rev 2
– Fixed Stemma connector pinned backward
– R3 set to 10k Ohm. Works fine and reduces unique part count.
– Updated Silkscreen & added QR code

Mounting Holes

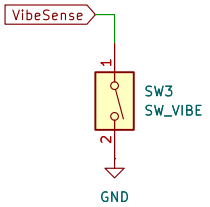


Power Switch – Battery Connector



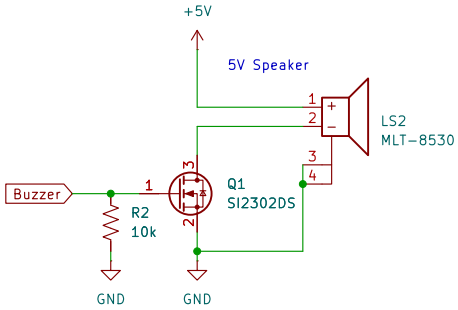
Note: This battery connector pinout seems to match standard adafruit batteries

Vibration Sensor

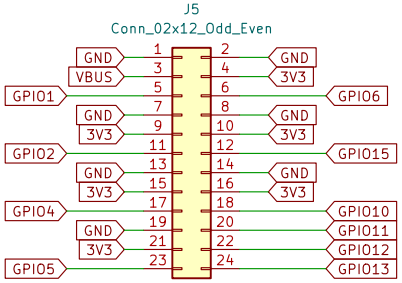


Note: Use GPIO as input with pull up to read the vibe sensor

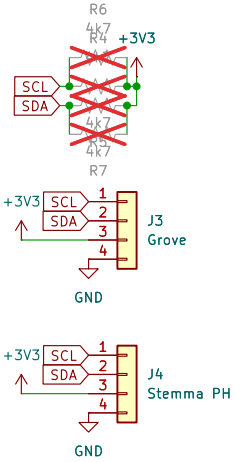
Speaker



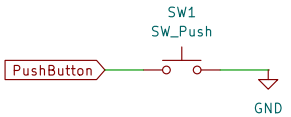
GPIO Header



I2C Connectors

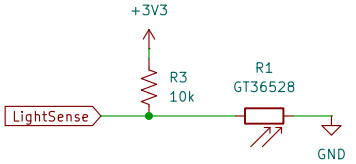


Push Button

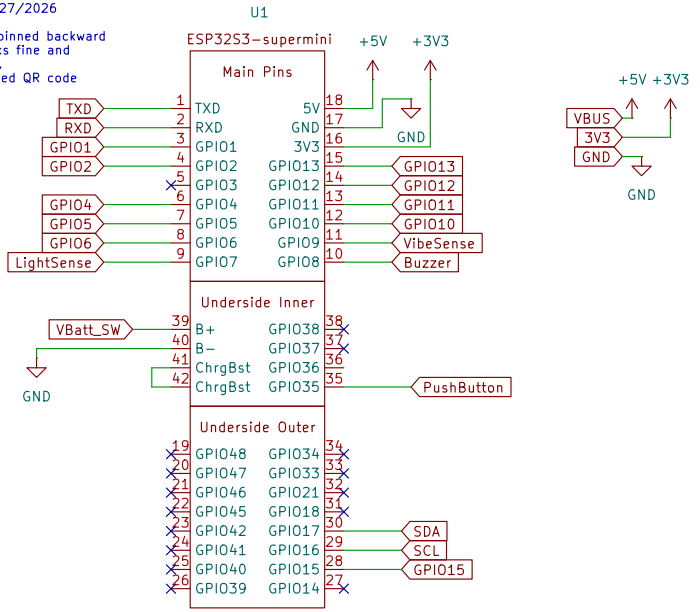


Note: Use GPIO as input with pull up to read the switch

Light Sensor



LDR is a CDS that measures:
– 500 ohms when I shine my flashlight on it
– 90k in a dim (hard to read) room



Tinkerfarm | WildCircuits | Brockport Original Builds

Sheet: /
File: Cobble–board.kicad_sch

Title: Cobble – ESP32S3–supermini dev board

Size: A4	Date: 2026–11–09	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 1/1